

Diffie-Hellman Key Exchange

Initiator

Responder

Pick secret a
Compute g^a

Pick secret b
Compute g^b

Exchange Public Keys

$g^a \longleftrightarrow g^b$

Compute $(g^b)^a$
 $= g^{ab}$

Compute $(g^a)^b$
 $= g^{ab}$

Symmetric Computation

Both parties compute g^{ab} independently

✓ Non-interactive (no round-trip needed)

✓ Roles are interchangeable

Key Encapsulation Mechanism

Initiator

Responder

$(ek, dk) \leftarrow \text{KeyGen}()$
Public: ek , Private: dk

Encapsulation Key

ek (1184 bytes for ML-KEM-768)

Ciphertext

ct (1088 bytes for ML-KEM-768)

$(ss, ct) \leftarrow \text{Encaps}(ek)$
Obtains shared secret ss

$ss \leftarrow \text{Decaps}(dk, ct)$
Obtains shared secret ss

Asymmetric Computation

Only dk holder can recover ss from ct

× Requires message exchange (round-trip)

× Roles are NOT interchangeable